

Tom S. Bertalan

APPLIED SCIENTIST IN DIFFERENTIABLE COMPUTING, PROBABILISTIC INFERENCE, AND NEUROSymbolic PROGRAM SYNTHESIS; APPLIED TO COMPLEX SCIENTIFIC AND ENGINEERING SYSTEMS.

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EDUCATION & TRAINING

• Johns Hopkins University Postdoctoral Fellow	2020 - 2024 <i>Chemical and Biological Engr.</i>	• Princeton University Ph.D. & Masters, Chemical and Biological Engr.	2012 - 2018
• The Massachusetts Institute of Technology Postdoctoral Associate	2018 - 2020 <i>Mechanical Engr.</i>	• The University of Alabama Bachelor of Science in Chemical and Biological Engr.; Minor in Math	2008 - 2012

PROFESSIONAL EXPERIENCE AND RESEARCH AREAS

Amgen

Cambridge, MA

DATA SCIENTIST

11/2024-Present

• **Mechanistic Modeling of Lyophilization and Filling**

- Designed and advocated a real-time lyophilization humidity digital twin using sequential Bayesian state estimation.
- Conceived, implemented, and released company-wide a data-validated mechanistic simulation of lyophilization.
- Calibrated model parameters using experimental data from seven production programs and designed a modular parameterization strategy to support pooled multi-product calibration workflows.
- Contributed to a growing first-principles mechanistic model of peristaltic filling, used by a broad internal customer base.
- Implemented pipelines to ingest high-volume accelerometer and flow data into data lake and co-designed a new product for the semantic data fabric.
- Specified data schemas for a clean library interface, supporting connected and interoperable digital twin architectures across bioprocess operations, and rapid GUI generation on EC2-backed Plotly Dash platform.
- Led five agile sprint for demos for stakeholder requirement capture, and executed three iterative deliveries company-wide of containerized and versioned package releases.

• **Hybrid Mechanistic+Black-box Modeling of CHO Cell Metabolism**

- Evaluated vendor offerings for Gaussian process-based time series system identification tools.
- Advised on efficient gradient-based training of neural differential equations with embedded optimality constraints, including custom adjoint sensitivity methods for memory efficiency.

• **Agile Development Automation Initiative**

- Ingested handwritten notes using a multimodal large-language model on Databricks.
- Accelerated sprint planning using the GitLab and Microsoft Graph APIs.

University of Massachusetts

Lowell, MA

RESEARCH SOFTWARE ENGINEER

1/2024-11/2024

• **Neurosymbolic System Identification of Dynamical Systems**

- Developed a neurosymbolic framework combining grammatical evolution for modular symbolic program synthesis with neural differential equations, enabling structured reasoning over scientific systems.
- Ensured cross-domain modularity between robotics and bioprocessing through a domain-specific language design in ANTLR.
- Outlined a future pipeline for LLM-assisted symbolic model assembly, integrating code generation and reasoning tools with differentiable physical simulators.

• **Model Predictive Control of CHO Bioreactor**

- Collected targeted experimental data from a medium-scale bioreactor and developed an online data-driven model-predictive controller with Aspen DMC3 to maximize economic factors, deploying to actual wet-lab hardware.
- Supervised a rigorous implementation of a SBML-encoded bioreactor digital twin, including a sequential-Bayes soft-sensor for latent bioreactor glucose concentration.
- Secured grant as PI to develop a central data lake for an NIH-seeded academic/industry consortium.

Johns Hopkins University

Baltimore, MD

POSTDOCTORAL FELLOW

2/2020-1/2024

• **Neural Differential Equations and Time-Series**

- Created a GUI for Bayesian experimental design; mentored team members on its use and maintenance.
- Improved accuracy of backpropagation and convergence time of PyTorch neural stochastic differential equation training.
- Advanced differentiable computing for learning continuous scientific laws from time-series data, bridging symbolic and statistical modeling through neural ODEs, PDEs, and SDEs; resulting in 9 publications and 10 conference presentations.
- Cut RNN inference burn-in from 25 to 5 samples using manifold learning.
- Led a team of biophysics and ML experts in creating a suite of simulation tools for CHO metabolism, including custom gradients for nets with constraints.

• **Robotic Perception and Control**

- Developed a variational autoencoder for end-to-end robotic localization.
- Trained a U-net on both open and custom datasets for real-time (>10hz) on-board semantic segmentation of drivable space.

The Massachusetts Institute of Technology

POSTDOCTORAL ASSOCIATE

Cambridge, MA

3/2018-2/2020

• **Nonlinear dynamics in neuroscience and bioprocessing**

- Wrote object-oriented library for fine- and coarse-grained simulations of neuronal dynamics, probing bifurcation and resonance behavior of a mammalian circadian rhythm model.
- Coauthored successful application for a \$1.8M NIH-led industry-academic partnership grant for a new bioreactor simulation.

• **Autonomous Vehicle Design and Pathfinding**

- Developed a model AV with firmware-level speed sensing and control commanded by checksummed bus communications.
- Simulated a comfort-optimized (jerk-minimizing) path planner with two-lane-ahead planning at 47 mph, and implemented a model-predictive path follower achieving 67 ms latency (IPOPT, CppAD) in C++ for real-time application.

Princeton University

NSF RESEARCH ASSISTANT

Princeton, NJ

9/2012-3/2018

Built a rover for Bayesian particle-filter SLAM with LIDAR using Gazebo and ROS, and a custom library for visualizing OpenCV pipelines and dynamic pruning of computational graphs. Simulated thousands of neurons in Numpy and MATLAB, and social agents in OpenMP-accelerated C++. Advised undergrads on theses ranging from power-grid opt. with Gurobi to combinatorial basis extraction.

SKILLS AND INTERESTS

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| • Tensorflow/Pytorch | • Grammatical evolution | • Retrieval-augmented generation (RAG) |
| • LangChain and HuggingFace Transformers | for code generation | • SQL(alchemy), HTML, CSS, JS, Flask, LAMP |
| • Databricks for data lakes and ML | • Bayesian inference | • Classical and DL-based computer vision |
| • Python, C++, Java | and probabilistic computing | • Trainee and peer mentoring |
| • NumPy, SciPy, SymPy, Matplotlib, SKLearn | • HPC with SLURM and OpenMP, and CUDA | • Home automation with Arduino, |
| • Linux, shell scripting, and git | • Differentiable physics for ML | Raspberry Pi, and 3D printing |
| • OPC-UA, MODBUS/serial, and ROS | • Mechanistic+data-driven | • Solo and orchestral violin performance |
| • Online optimization (IPOPT, CppAD, CasADi) | grey-box modeling | • Windsurfing and small-boat sailing |

SELECTED PUBLICATIONS

Implementation and (Inverse Modified) Error Analysis for implicitly-templated ODE nets

A. Zhu, B. Zhu, [Tom Bertalan](#), Y. Tang, I. Kevrekidis

2024

SIAMDS

Data-driven and Physics Informed Modelling of Chinese Hamster Ovary (CHO) Cell Bioreactors

T. Cui, [Tom Bertalan](#), N. Ndahiro, P. Khare, M. Betenbaugh, C. Maranas, I. Kevrekidis

2024

Comp. & Chem. Engr.

Symbolic regression and modular neural diff. eq. for bioproc. engr. and robotics

[Tom Bertalan](#), Jaeweon Lee, Zhao Wang, Seongkyu Yoon, I. Kevrekidis

2024

MLDS 4

Certified Invertibility in Neural Networks via Mixed-Integer Programming

T. Cui, [Tom Bertalan](#), G. Pappas, M. Morari, I. Kevrekidis, M. Fazlyab

2023

L4DC 2023 — PMLR

Learning effective stochastic differential equations from microscopic simulations: linking stochastic numerics to deep learning

F. Dietrich, A. Makeev, G. Kevrekidis, N. Evangelou, [Tom Bertalan](#), S. Reich, I. Kevrekidis

2023

Chaos

Personalized Algorithm Generation: A Case Study in Meta-Learning ODE Integrators

Y. Guo, F. Dietrich, [Tom Bertalan](#), D. Doncevic, M. Dahmen, I. Kevrekidis, Q. Li

2022

SIAM J. Sci. Comp.

Learning emergent PDEs in a learned emergent space

F. Kemeth, [Tom Bertalan](#), T. Thiem, S. Moon, C. Laing, I. Kevrekidis

2022

Nature Comm.

Local conformal autoencoder for standardized data coordinates

E. Peterfreund, O. Lindenbaum, F. Dietrich, [Tom Bertalan](#), M. Gavish, I. Kevrekidis, R. Coifman

2020

PNAS

On Learning Hamiltonian Systems from Data

[Tom Bertalan](#), F. Dietrich, I. Mezic, and I. Kevrekidis

2019

Chaos

PUBLICATIONS

Comparing Kinetic versus Stoichiometric Priorities in Hybrid Models of CHO Metabolism Tom Bertalan, P. Khare, M. Betenbaugh, I. Kevrekidis	2025 (under review) NPJ Sys. Bio. and Appl.
Integ. of Bayesian Optim. and Solution Thermo. to Opt. Media Design for Mammalian Biomfg. N. Ndahiro, E. Ma, Tom Bertalan, M. Donohue, Y. Kevrekidis, and M. Betenbaugh	2025 iScience
Machine Learning Approaches to Problem Well-Posedness Tom Bertalan, G. Kevrekidis, E. Rebrova, S. Mishra, I. Kevrekidis	In Preparation
Data-driven and Physics Informed Modelling of Chinese Hamster Ovary (CHO) Cell Bioreactors T. Cui, Tom Bertalan, N. Ndahiro, P. Khare, M. Betenbaugh, C. Maranas, I. Kevrekidis	2024 Comp. & Chem. Engr.
Implementation and (Inverse Modified) Error Analysis for implicitly-templated ODE nets A. Zhu, B. Zhu, Tom Bertalan, Y. Tang, I. Kevrekidis	2024 SIAMDS
Transf. establishing equiv. across neural nets.: when have two nets. learned the same task? Tom Bertalan, F. Dietrich, I. Kevrekidis	2024 Chaos
Certified Invertibility in Neural Networks via Mixed-Integer Programming T. Cui, Tom Bertalan, G. Pappas, M. Morari, I. Kevrekidis, M. Fazlyab	2023 L4DC 2023 — PMLR
Some of the variables, some of the parameters, some of the times, with some things known: Identification with partial information S. Malani, Tom Bertalan, T. Cui, M. Betenbaugh, J. Avalos, I. Kevrekidis	2023 Comp. & Chem. Engr.
Learning effective stochastic differential equations from microscopic simulations: linking stochastic numerics to deep learning F. Dietrich, A. Makeev, G. Kevrekidis, N. Evangelou, Tom Bertalan, S. Reich, I. Kevrekidis	2023 Chaos
Learning emergent PDEs in a learned emergent space F. Kemeth, Tom Bertalan, T. Thiem, S. Moon, C. Laing, I. Kevrekidis	2022 Nature Comm.
Personalized Algorithm Generation: A Case Study in Meta-Learning ODE Integrators Y. Guo, F. Dietrich, Tom Bertalan, D. Doncevic, M. Dahmen, I. Kevrekidis, Q. Li	2022 SIAM J. Sci. Comp.
Initializing LSTM internal states via manifold learning F. Kemeth, Tom Bertalan, N. Evangelou, T. Cui, S. Malani, I. Kevrekidis	2021 Chaos
Development of closures for coarse-scale modeling of multiphase and free surface flows using machine learning C. Linares, Tom Bertalan, E. Koronaki, Jicai Lu, G. Tryggvason, I. Kevrekidis	2021 APS Bulletin
Global and local reduced models for interacting, heterogeneous agents T. Thiem, F. Kemeth, Tom Bertalan, Carlo Liang, I. Kevrekidis	2021 Chaos
Local conformal autoencoder for standardized data coordinates E. Peterfreund, O. Lindenbaum, F. Dietrich, Tom Bertalan, M. Gavish, I. Kevrekidis, R. Coifman	2020 PNAS
Emergent spaces for coupled oscillators T. Thiem, M. Kooshkbaghi, Tom Bertalan, Carol Laing, I. Kevrekidis	2020 Front. in Comp. Neuro.
On Learning Hamiltonian Systems from Data Tom Bertalan, F. Dietrich, I. Mezic, and I. Kevrekidis	2019 Chaos
An Emergent Space for Distributed Data with Hidden Internal Order through Manifold Learning F. Kemeth, Sindre Haugland, F. Dietrich, Tom Bertalan, Kevin Höhle, Q. Li, Erik Bollt, Ronen Talmon, Katharina Krischer, and I. Kevrekidis	2017 IEEE Access
Coarse-grained descriptions of dynamics for networks with both intrinsic and structural heterogeneities Tom Bertalan, W. Wu, C. Laing, C. William Gear, and I. Kevrekidis.	2017 Front. in Comp. Neuro.

Dimension reduction in heterogeneous neural networks: Generalized Polynomial Chaos (gPC) and ANalysis-Of-Variance (ANOVA)

M. Choi, [Tom Bertalan](#), C. Laing, and I. Kevrekidis.

2016
Euro. Phys. J., Special Topics

OpenMG: a new multigrid implementation in Python

[Tom Bertalan](#), A. Islam, R. Sidje, and E. Carlson

2014
Num. Lin. Alg. with App.

PRESENTATIONS

Integrated Modeling and Optim. of Secondary Drying for Pharma. Freeze-Drying Proc.

[Tom Bertalan](#), M. Maritato, A. Ganguly, W. Al Bakri, R. Tukra, B. Lacetti

2025
AIChE 2025 (poster)

Integrated Continuous USP Platform for Max. Productivity and Closed-Loop Controlled CQA

J. Lee, [Tom Bertalan](#), Z. Wang, Y. Choi, S. Song, H. Yamanaka, A. Rajendran, J. Han, S. Yoon

2025
AIChE 2025

Digital Twin-Enabled Control of Productivity and CQA in CHO Cell Cultures

Y. Choi, S. Yoon, Z. Wang, [Tom Bertalan](#), J. Lee

2025
AIChE 2025 (poster)

Beyond Trial-and-Error: An in silico Model for Accelerated Peristaltic Fill Recipe Development

Maritato, Ingram, Sao, Heymann, [Tom Bertalan](#), Banerjee, Icten Gencer, Ganguly, Pearson, Schlegel

2025
AIChE 2025 (poster)

Symbolic regression and modular neural diff. eq. for bioproc. engr. and robotics

[Tom Bertalan](#), Jaeweon Lee, Zhao Wang, Seongkyu Yoon, I. Kevrekidis

2024
MLDS 4

Certified Invertibility in Neural Networks via Mixed-Integer Programming

T. Cui, [Tom Bertalan](#), G. Pappas, M. Morari, I. Kevrekidis, M. Fazlyab

2023
Learning for Dyn. Sys.

Coarse-grained and emergent distributed-parameter systems from data

Hassan Arbabi, F. Kemeth, [Tom Bertalan](#), I. Kevrekidis

2021
American Control Conf.

Data-driven model reduction and discovery

T. Thiem, [Tom Bertalan](#), F. Kemeth, Yorgos Psarellis, I. Kevrekidis

2020
AIChE

Dynamical-systems-guided learning of PDEs from data

Hassan Arbabi, [Tom Bertalan](#), A. Roberts, I. Kevrekidis

2020
AIChE

On the data-driven discovery and calibration of closures

S. Lee, Yorgos Psarellis, C. Siettos, [Tom Bertalan](#), D. Amchin, T. Bhattacharjee, S. Datta, I. Kevrekidis

2020
AIChE

Connections between residual networks and explicit numerical integrators, and applications to identification of noninvertible dynamical systems

T. Cui, [Tom Bertalan](#), Yorgos Psarellis, I. Kevrekidis

2020
AIChE

Neural network approach to reduced order modeling of multiphase flows

C. Linares, [Tom Bertalan](#), S. Lee, J. Lu, G. Tryggvason, I. Kevrekidis

2020
APS Div. of Fluid Dyn.

PDE+PINN: Learning and Solving a PDE at the Same Time

[Tom Bertalan](#), F. Kemeth, T. Cui, I. Kevrekidis

2020
AIChE

Learning Partial Differential Equations from Discrete Space Time Data: Convolutional and Recurrent Networks, and Their Relations to Traditional Numerical Methods

[Tom Bertalan](#), F. Dietrich, T. Thiem, R. Farber, I. Kevrekidis, A. Roberts

2019
AIChE

Recurrent Neural Networks, Numerical Integrators and Nonlinear System Identification

[Tom Bertalan](#), R. Farber, T. Thiem, F. Dietrich, I. Kevrekidis

2018
AIChE

Coarse-Scale PDEs from Microscopic Observations Via Machine Learning

S. Lee, M. Kooshkbaghi, C. Siettos, [Tom Bertalan](#), and I. Kevrekidis

2019
AIChE

When Have Two Networks Learned the Same Task? Data-Driven Transformations between System Representations

[Tom Bertalan](#), F. Dietrich, T. Thiem, I. Kevrekidis

2019
AIChE

Coarse modeling of circadian rhythms in heterogeneous neural networks	2017; 2016
Tom Bertalan , C. William Gear, Yannis G. Kevrekidis, M. Henson, E. Herzog, and C. Laing	<i>Dyn. Days 2017; AIChE</i>
Coarse-graining of heterogeneous neural dynamics	2015
Tom Bertalan , M. Choi, C. Laing, I. Kevrekidis	<i>AIChE</i>
Heterogeneity and reduction for complex network dynamics	2014
I. Kevrekidis, A. Holiday, Tom Bertalan , and C. Laing	<i>AIChE</i>
Coarse-graining Network Dynamics	2013
A. Holiday and Tom Bertalan	<i>Network Front.</i>
nSpyres: an open-source, Python-based framework for simulation of flow through porous media	2012
E. Carlson, A. Islam, F. Dumkwu, and Tom Bertalan	<i>Interpore 2012</i>
OpenMG: a new multigrid implementation in Python	2012
Tom Bertalan , A. Islam, R. Sidje, and E. Carlson	<i>Proc. 11th Python in Sci. Conf.</i>
ESIM: a framework for simulation of dominance hierarchy formation in small animal groups	2012
Tom Bertalan and R. Earley	<i>UA Hon. Undergr. Res. Conf.</i>
An open-source computing cluster for virtual experiments with variable parameters	2011
Tom Bertalan and E. Carlson	<i>UA Hon. Undergr. Res. Conf.</i>